Lecture 2

Some problems

1. $\lim_{x\to 0} \frac{x}{\sqrt{x+1}-1}$ $\left[\frac{0}{0}\right]$

$\lim_{x\to 0} \frac{x(\sqrt{x+1}+1)}{(\sqrt{x+1}-1)(\sqrt{x+1}+1)}$

$\lim_{x\to 0} \frac{x(\sqrt{x+1}+1)}{x+1-1}$

$\lim_{x\to 0} \frac{x(\sqrt{x+1}+1)}{x}$

$\lim_{x\to 0} (\sqrt{x+1}+1) = 2.$ ※

2. $\lim_{x\to 2} \frac{2x^2-5x+2}{5x^2-7x-6}$

Do yourself

3. $\lim_{x\to\infty} (\sqrt{x^6+5} - x^3)$

$\lim_{x\to\infty} (\sqrt{x^6+5} - x^3)\frac{(\sqrt{x^6+5}+x^3)}{(\sqrt{x^6+5}+x^3)}$

(~~Do yourself~~)

$\lim_{x\to\infty} \frac{x^6+5-x^6}{\sqrt{x^6+5}+x^3}$

$\lim_{x\to\infty} \frac{5}{\sqrt{x^6+5}+x^3}$

$\lim_{x\to\infty} \frac{5}{x^3(\sqrt{1+5/x^3}+1)}$

$\lim_{x\to\infty} \frac{5/x^3}{\sqrt{1+5/x^3}+1} = 0$

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<u>Example:</u> Given that

$$f(x) = \begin{cases} \frac{1}{x+2} & , x < -2 \\ x^2 - 5 & , -2 \leq x \leq 3 \\ \sqrt{x+13} & , x > 3 \end{cases}$$

Find the limit $\lim_{x \to -2} f(x)$ and $\lim_{x \to 3} f(x)$ if it exist ?

Solⁿ. <u>at $x = -2$</u>

$$L.H.L = \lim_{x \to -2^-} f(x) = \lim_{x \to -2^-} \frac{1}{x+2}$$

$$= -\infty$$

$$R.H.L. = \lim_{x \to -2^+} f(x) = \lim_{x \to -2} x^2 - 5 = -1$$

So, $L.H.L \neq R.H.L.$ thus $\lim_{x \to -2} f(x)$ doesn't exist.

<u>2nd part</u>

<u>At $x = 3$</u>

$$L.H.L = \lim_{x \to 3^-} f(x) = \lim_{x \to 3} x^2 - 5 = 4$$

$$R.H.L. = \lim_{x \to 3^+} f(x) = \lim_{x \to 3} \sqrt{x+13} = \sqrt{16} = 4$$

So, $L.H.L. = R.H.L$ and $\lim_{x \to 3} f(x) = 4$ at $x = 3$.

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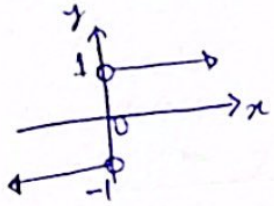
Problem: ~~Find the lim~~

Estimate the limit: $\lim_{x \to 0} \frac{x}{|x|}$ if it exist, ~~where~~ and $x \neq 0$

Sol^n: Assume that



$$f(x) = \frac{x}{|x|} = \begin{cases} \frac{x}{x} = 1 & , x > 0 \\ \frac{x}{-x} = -1 & , x < 0 \end{cases}$$

$$\text{L.H.L.} = \lim_{x \to 0^-} f(x) = \lim_{x \to 0} \frac{x}{-x} = \lim_{x \to 0} -1 = -1$$

$$\text{R.H.L.} = \lim_{x \to 0^+} f(x) = \lim_{x \to 0} 1 = 1$$

Thus, L.H.L. $\neq$ R.H.L. . Therefore, $\lim_{x \to 0} \frac{x}{|x|}$ doesn't exist.

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The Squeez Theorem: Let f, g, and h be functions satisfying

$$\boxed{g(x) \leq f(x) \leq h(x)}$$

for all x in some open interval containing the number c, with the possible exception that the inequalities need not hold at c. If g and h have the same limit as x approaches c, say

$$\boxed{\lim_{x \to c} g(x) = \lim_{x \to c} h(x) = L}$$

then f also has this limit as x approaches c, that is

$$\boxed{\lim_{x \to c} f(x) = L.}$$

Problem: Estimate: $\lim_{x \to 0} x \sin\left(\frac{1}{x}\right)$ if exists?.

Sol^n: Assume, $f(x) = x \sin\left(\frac{1}{x}\right)$

Since $-1 \leq \sin \frac{1}{x} \leq 1$

$$-|x| \leq |x| \sin \frac{1}{x} \leq |x| \Rightarrow \lim_{x \to 0} -|x| \leq \lim_{x \to 0} |x| \sin x \leq \lim_{x \to 0} |x|$$

$\sin |x| \to 0$ as $x \to 0$ i.e. $\lim_{x \to 0} |x| = 0$

Applying squeez theorem,

$$\lim_{x \to 0} x \sin\left(\frac{1}{x}\right) = 0$$

(* Why we don't use $\lim_{x \to 0} x \cdot \lim_{x \to 0} \sin\left(\frac{1}{x}\right) = 0$?)

TRY YOURSELF: $g(x) = \lim_{x \to 0} x \cos(1/x)$

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Our previous example $g(x) = \begin{cases} \dfrac{x^2+4x-12}{x^2-2x} & \text{if } x \neq 2 \\ 6 & \text{if } x = 2 \end{cases}$

We see that Left hand limit and Right hand limit exist and $\lim_{x \to 2} g(x) = 4$ but functional value $f(2) = 6$.

So, we have now get some extra condition for ~~some~~ one kind of function and which called continuous at $x = a$.

<u>Definition of Continuity:</u>

A function f is said to be continuous at $x = a$ provided the following conditions are satisfied:

1. $f(a)$ is defined
2. $\lim_{x \to a} f(x)$ exists

and 3. $\lim_{x \to a} f(x) = f(a)$.

<u>Example:</u> Given $f(x) = \begin{cases} x^2+1, & x > 0 \\ 1, & x = 0 \\ x+1, & x < 0 \end{cases}$.

Find the $\lim_{x \to 0} f(x)$. ~~For which~~ Show that the function $f(x)$ is continuous at $x = 0$.

<u>Sol^n:</u>

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (1+x) = 1$$

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$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0} (x^2+1) = 1 \quad \text{limit exists.}$$

and $f(0) = 1$

Thus, $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = 1 = f(0)$.

Therefore, $f(x)$ is continuous at $x = 0$. ※

<u>Example:</u> Given that

$$f(x) = \begin{cases} \dfrac{e^{1/x^2}}{e^{1/x^2} - 1} & , \text{if } x \neq 0 \\ 1 & , \text{if } x = 0 \end{cases}$$

Does $f(x)$ continuous at $x = 0$?

<u>Sol</u>.

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Properties of continuity

If the function f and g are continuous at c, then

ⓐ $f \pm g$ is continuous at c.

ⓑ fg is continuous at c.

ⓒ f/g is continuous at c if $g(c) \neq 0$ and has a discontinuity at c if $g(c) = 0$.

** If $\lim_{x \to c} g(x) = L$ and if f is continuous at a, then

$$\lim_{x \to c} f(g(x)) = f\left(\lim_{x \to c} g(x)\right).$$

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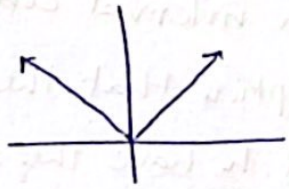
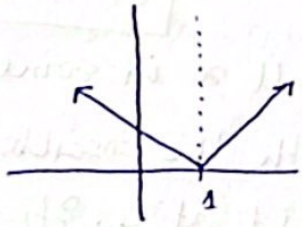
Prop Problem: Given that $f(x) = |x| + |x-1|$. Does it continuous at $x = 0$ and at $x = 1$?

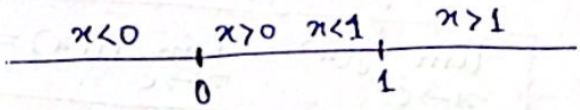
$$|x-1| = \begin{cases} x-1, & x-1>0 \\ & x>1 \\ -(x-1), & x-1<0 \\ & x<1 \end{cases}$$

Sol^n



$|x|$





$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (|x| + |x-1|) = \lim_{x\to 0} [x - (x-1)] = \lim_{x\to 0} (-2x+1) = 1$$

$$\lim_{x\to 0^+, 1^-} f(x) = \lim_{x\to 0,1} [x-(x-1)] = \lim_{x\to 0,1} 1 = 1$$

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1} [x+(x-1)] = \lim_{x\to 1} (2x-1) = 1$$

Therefore $f(x)$ is continuous at $x=0$ and at $x=1$.

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Some extra problems:

1. Find the limit if it exists, otherwise explain why it does not exist:

(a) $\lim_{x \to -3} \frac{|x^2-9|(x+2)}{x^2+7x+12}$.

(b) $\lim_{x \to 4} \frac{16-x^2}{x^3-5x^2+4x}$.

(c) $\lim_{x \to 0} \frac{x}{3-\sqrt{x+9}}$.

(d) $\lim_{x \to -3} \frac{\sqrt{2x+22}-4}{x+3}$.

(e) $\lim_{t \to -1} \frac{t+1}{|t+1|}$.

(f) $\lim_{x \to 0} x^4 \sin\left(\frac{\pi}{x}\right)$, $x \neq 0$.

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